

Problem Solving and Search Algorithms

Introduction to AI problem-solving paradigms,

Problem formulation and search space exploration,

Uninformed search algorithms: depth-first search, breadth-first search,

Informed search algorithms: A* search, Domain relaxations and heuristic functions

Local Search and Evolutionary Algorithms: Hill climbing and simulated annealing, Genetic algorithms and genetic programming

Adversarial Search: Minimax algorithm, Alpha-beta pruning and game tree search Advanced Search Algorithms: iterative deepening,

bidirectional search 10 30% 2 Game Playing: Overview, And Example Domain: Overview, MiniMax, Alpha-Beta Cut-off, Refinements, Iterative deepening

There are two methods for traversing graphs, which are as follows:

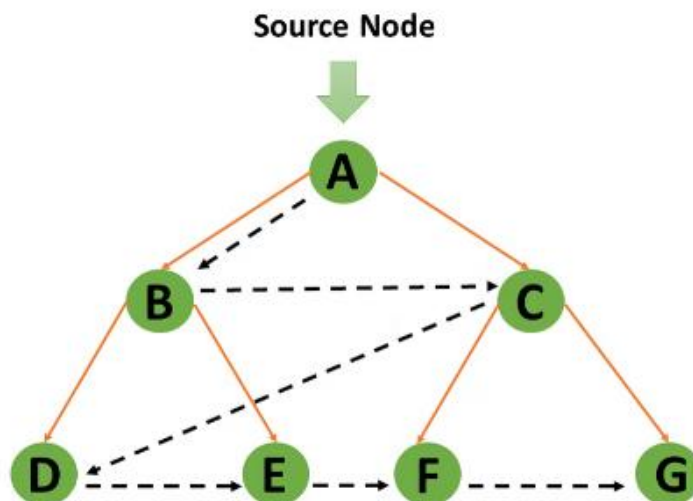
- Breadth-First Search or BFS Algorithm
- Depth-First Search or DFS Algorithm

Breadth-First Search Algorithm or BFS is the most widely utilized method.

BFS is a graph traversal approach in which you start at a source node and layer by layer through the graph, analyzing the nodes directly related to the source node. Then, in BFS traversal, you must move on to the next-level neighbor nodes.

According to the BFS, you must traverse the graph in a breadthwise direction:

- To begin, move horizontally and visit all the current layer's nodes.
- Continue to the next layer.



Complexity Of Breadth-First Search Algorithm

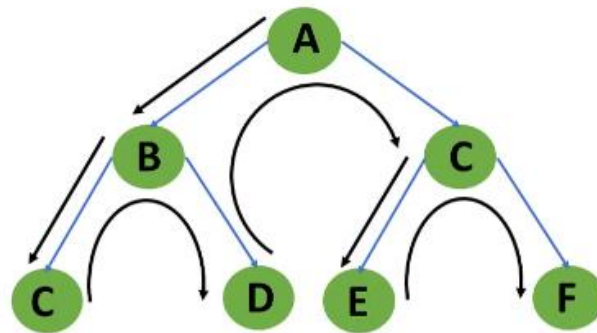
The time complexity of the breadth-first search algorithm : The time complexity of the breadth-first search algorithm can be stated as $O(|V|+|E|)$ because, in the worst case, it will explore every vertex and edge. The number of vertices in the graph is $|V|$, while the edges are $|E|$.

2. What is the BFS algorithm used for?

- The BFS algorithm is used in Crawlers, which is used by search engines.
- The BFS algorithm is used in peer-to-peer networks.
- The BFS algorithm is used in a GPS navigation system.
- The BFS algorithm is used in the broadcasting network.

What is a Depth-First Search Algorithm?

The depth-first search or DFS algorithm traverses or explores [data structures](#), such as [trees](#) and [graphs](#). The [algorithm](#) starts at the root node (in the case of a graph, you can use any random node as the root node) and examines each branch as far as possible before backtracking.



Hill climbing has three well-known drawbacks:

1. **Local Maxima:** A local maximum is a state that is better than all its neighbours but is not better than some other states further away.
2. **Plateau:** A plateau is a flat area of the search space in which, a whole set of neighbouring states have the same values.
3. **Ridge:** Ridge is a special kind of local maximum. It is an area of the search space that is higher than surrounding areas and that itself has slop.

